Day: M T W T F S

Date: ______

Mixed data :- (Feature engineering) :-

Mixed

↓

e.g:- a Colo having value like

B5
C23
D41

B5 → Catogorical, Numerical.

Cat	Num
B	5
C	23
D	21

or

7
d
2
B

So make a coloumn in which put numerical values and to catogorical put NA.

7
4
NA
3
NA

OR

Make 2 separate tables in which one coloumn has NA & other has numerical values.

These 2 are techniques for handling mixed data.

Date: ______ Day: M T W T F S

Handling Date & Time variable:-

date

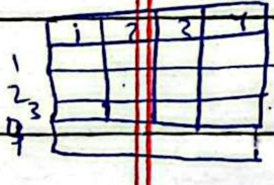
23-Aug-2012

day month Year day of week quarter

time 08:05:52.

Here much information is hidden, we can extract it.

Handling missing data:



Missing value

Remove them. Impute them fill.

Univariate Multivariate

Numerical Cat King impurator. Iterative

Date:____________ Day: M T W T F S

Complete case analysis :-

Removing missing values, that approach is known as ....

- Discarding rows where values in any of coloumn is missing.

Assumptions for CCA :-

1) Missing completely at random

500 rows, 4 coloumns.

Age 50 → 950 , 4

$\boxed{\text{MCAR}}$

Advantages :-

1) Easy to implement

2)

Disadvantages :-

1) Exclude a large fraction of original dataset.

2) Excluded observation

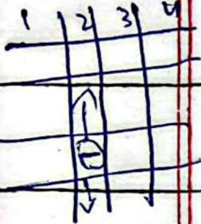
3) When using our models, model will how to handle missing data

When to Use CCA :-

1) MCAR

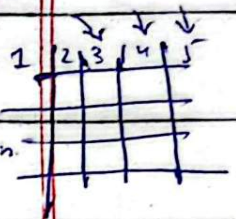
2) 5% < missing data.

Date: Day: M T W T F S



Numerical data.

- Univariate
- Multivariate imputation.



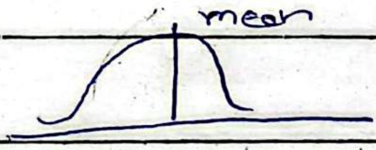
Univariate
- mean/median
- Arbitrary
- End of distribution
- Random

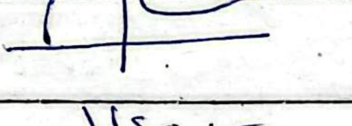
a) Mean/Median imputation:-

Age

Age
27
32
NA
29

→ Mean/Median.





Benefits:-
1) Simple
2)

Disadvantages
1) Distribut shape.
2) Outliers.
3) Corelation charges.

When to Use:-
1) MCAR.
2) less then 5% missing.

b) Arbitrary value imputation:-

↳ Cat → NA → Missing.

Num → -1, 0, 99.99

age → 99, -1

Benefits:-

1) easy to apply

disadvant:-

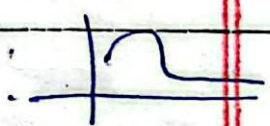
1) PDF distort.

2) Variance change.

3) Covariance ″.

- Data is not missing at random.

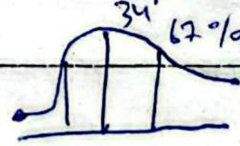
c) End of distribution imputation:-



→ In arbitrary we replace missing value with arbitrary no.

→ (99) →

↳ End of dist



Normally dist

↳ (Mean + 3σ)

(mean − 3σ)

Skewed [IQR Proximity rule

↳ $Q_1 - 1.5\,IQR$

$Q_3 + 1.5\,IQR$

$\boxed{Q_3 - Q_1}$

Date: Day: M T W T F S

25 % Q_1 percentile.

75 % Q_3

- In arbitrary we replace missing value with any no.
- In end of dist we take extreme values.
- In both we do importance of missing values.

Benefit:-

- easy to use. → Same as Arbitrary.